Akshai Narayan Srinivasan

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Education

University of Washington Seattle, WA
Dual Degree: BS in Computer Engineering, BS in Applied Mathematics *June 2027*

Relevant Coursework: Machine Learning, Systems Programming, Digital Design, Data Structures & Parallelism, Hardware Software Interface, Discrete Mathematics & Probability, Dynamical Systems, Applied Linear Algebra

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, Dart
Cloud & Databases: AWS (Lambda, API Gateway, S3, DynamoDB, Bedrock), PostgreSQL, MongoDB, MySQL
Frameworks & Tools: ROS2, PyTorch, Isaac Lab, FastAPI, Claude API, Flask, WebSockets, Next.js, Stanford CoreNLP, Git

Experience

General Motors: Cruise Sunnyvale, CA
Machine Learning Engineering Intern *June 2026 - Present*

- Developing **autonomous vehicle** infrastructure for a web-based platform visualizing multi-source **ROS data streams**
- Accelerating **machine learning workflows** by optimizing spatial data rendering tools used to validate perception and localization model outputs
- Optimizing **3D panel data hierarchy** interface to streamline how engineers navigate, filter, and render dense spatial datasets and real-time sensor logs

Washington Embodied Intelligence and Robotics Development Lab Seattle, WA
Undergraduate Research Assistant *Feb 2026 – June 2026*

- Developed sim-to-real **reinforcement learning** policies for complex robotic assembly tasks using **Isaac Lab** and **RSL-RL**, leveraging WandB for experiment tracking
- Tuned **PPO hyperparameters** and reward formulations to maximize dexterous manipulation success rates under aggressive **domain randomization** of object poses and physics dynamics
- Implemented **linear curriculum** learning, **ADR**, and **LSTM-based policy architectures** to mitigate reality gaps and maximize sim-to-real transfer robustness

Neptune Medical Inc. Burlingame, CA
Robotic Software Engineering Intern *Jun 2025 – Sep 2025*

- Developed controls software for **robotic endoscope** using **ROS** and **Python**, orchestrating communication between controllers, sensors, and actuators
- Built **C++ fault detection framework** for haptic input device with automatic failsafe protocols, preventing unsafe robot operations during **live procedures**
- Designed **tendon saturation algorithm** with integrated **haptic feedback** that dynamically generates resistive forces when tendons reach operational limits, preventing mechanical damage to endoscope

DubHacks Seattle, WA
Director of Technology *Jan 2025 – Present*

- Overseeing all software development and infrastructure for the **largest collegiate hackathon** in the Pacific Northwest **(1,200+ participants)** using **FastAPI, Next.js, TypeScript, and React**
- Engineered high-performance judging platform with **OAuth2/JWT authorization**, leveraging **WebSockets** and event-driven architecture to achieve **sub-100ms** latency for 150+ concurrent sessions
- Architected scalable **MongoDB** cluster handling **376K edge requests** and **122K serverless function** invocations during peak hackathon traffic with **100% uptime** and zero data loss

Sample Projects (Portfolio: akshaisrin.live)

DuckRL | *PyTorch, MuJoCo, Reinforcement Learning, PPO* | https://akshaisrin.live/DuckRLWebsite June 2026

- Investigated sim-to-real transfer for **reinforcement learning**-based bipedal locomotion via **Rapid Motor Adaptation (RMA)** using **Proximal Policy Optimization (PPO), PyTorch, and MuJoCo Sim**
- Ablated **domain randomization** across physics parameters (mass, friction, motor gain) using extrinsic-conditioned **MLP neural networks** to identify strategies driving **sim-to-real** robustness
- Stress-tested learned policies under **morphological perturbations** (link lengths, center of mass, foot geometry) to characterize failure modes of current DR pipelines